Lecture 12: Permutations & Combinations

Section 6.3 + 6.5

Section Summary

- Permutations
- Combinations
- Permutations with Repetitions
- Combinations with Repetitions

Permutations

Definition : A *permutation* of a set of distinct objects is an ordered arrangement of these objects. An ordered arrangement of r elements of a set is called an *r-permutation*.

Example : Let $S = \{1,2,3\}$.

- The ordered arrangement 3,1,2 is a permutation of S.
- The ordered arrangement 3,2 is a 2-permutation of S.

- The number of r-permutations of a set with n elements is denoted by $P(n,r)$.
  - The 2-permutations of $S = \{1,2,3\}$ are 1,2; 1,3; 2,1; 2,3; 3,1; and 3,2. Hence, $P(3,2) = 6$.

Solving Counting Problems by Counting Permutations (*continued*)

Example : Suppose that a saleswoman has to visit eight different cities. She must begin her trip in a specified city, but she can visit the other seven cities in any order she wishes. How many possible orders can the saleswoman use when visiting these cities?

Solution : The first city is chosen, and the rest are ordered arbitrarily. Hence the orders are:

$$7! = 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 5040$$

If she wants to find the tour with the shortest path that visits all the cities, she must consider 5040 paths!

A Formula for the Number of Permutations

Theorem : If n is a positive integer and r is an integer with $1 \leq r \leq n$, then there are

$$P(n, r) = n(n-1)(n-2)\cdots(n-r+1)$$

r-permutations of a set with n distinct elements.

Proof : Use the product rule. The first element can be chosen in n ways. The second in $n-1$ ways, and so on until there are $(n-(r-1))$ ways to choose the last element.

- Note that $P(n, 0) = 1$, since there is only one way to order zero elements.

Corollary : If n and r are integers with $1 \leq r \leq n$, then

$$P(n, r) = \frac{n!}{(n-r)!}$$

Solving Counting Problems by Counting Permutations

Example : How many ways are there to select a first-prize winner, a second prize winner, and a third-prize winner from 100 different people who have entered a contest?

Solution :

$$P(100,3) = 100 \cdot 99 \cdot 98 = 970{,}200$$

Solving Counting Problems by Counting Permutations (*continued*)

Example : How many permutations of the letters *ABCDEFGH* contain the string *ABC* ?

Solution : We solve this problem by counting the permutations of six objects, *ABC*, *D*, *E*, *F*, *G*, and *H*.

$$P(6, 6) = 6! = 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 720$$

Combinations

Definition : An *r-combination* of elements of a set is an unordered selection of *r* elements from the set. Thus, an *r*-combination is simply a subset of the set with *r* elements.

- The number of *r*-combinations of a set with n distinct elements is denoted by $C(n, r)$. The notation $\binom{n}{r}$ is also used and is called a *binomial coefficient*.

Example : Let *S* be the set $\{a, b, c, d\}$. Then $\{a, c, d\}$ is a 3-combination from S. It is the same as $\{d, c, a\}$ since the order listed does not matter.

- $C(4,2) = 6$ because the 2-combinations of $\{a, b, c, d\}$ are the six subsets $\{a, b\}$, $\{a, c\}$, $\{a, d\}$, $\{b, c\}$, $\{b, d\}$, and $\{c, d\}$.

Combinations

Theorem : The number of r-combinations of a set with n elements, where $n \geq r \geq 0$, equals

$$C(n,r) = \frac{n!}{(n-r)!r!}.$$

Proof: By the product rule $P(n, r) = C(n,r) \cdot P(r,r)$. Therefore,

$$C(n,r) = \frac{P(n,r)}{P(r,r)} = \frac{n!/(n-r)!}{r!/(r-r)!} = \frac{n!}{(n-r)!r!}.$$

Combinations

Example : How many poker hands of five cards can be dealt from a standard deck of 52 cards? Also, how many ways are there to select 47 cards from a deck of 52 cards?

Solution : Since the order in which the cards are dealt does not matter, the number of five card hands is:

$$C(52,5) = \frac{52!}{5!47!}$$
$$= \frac{52\cdot 51\cdot 50\cdot 49\cdot 48}{5\cdot 4\cdot 3\cdot 2\cdot 1} = 26\cdot 17\cdot 10\cdot 49\cdot 12 = 2,598,960$$

- The different ways to select 47 cards from 52 is

$$C(52,47) = \frac{52!}{47!5!} = C(52,5) = 2,598,960.$$

This is a special case of a general result. →

Combinations

Corollary : Let n and r be nonnegative integers with $r \leq n$. Then $\boldsymbol{C(n, r) = C(n, n - r)}$.

Proof : From Theorem 2, it follows that

$$C(n, r) = \frac{n!}{(n-r)!r!}$$

and $C(n, n-r) = \frac{n!}{(n-r)![n-(n-r)]!} = \frac{n!}{(n-r)!r!}$.

Hence, $C(n, r) = C(n, n - r)$.

This result can be proved without using algebraic manipulation. →

Combinations

Example : How many ways are there to select five players from a 10-member tennis team to make a trip to a match at another school.

Solution : The number of combinations is

$$C(10,5) = \frac{10!}{5!5!} = 252.$$

Example : A group of 30 people have been trained as astronauts to go on the first mission to Mars. How many ways are there to select a crew of six people to go on this mission?

Solution : The number of possible crews is

$$C(30,6) = \frac{30!}{6!24!} = \frac{30 \cdot 29 \cdot 28 \cdot 27 \cdot 26 \cdot 25}{6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1} = 593,775\ .$$

Permutations with Repetition

Theorem : The number of r-permutations of a set of n objects with repetition allowed is n^r.

Proof : There are n ways to select an element of the set for each of the r positions in the r-permutation when repetition is allowed. Hence, by the product rule there are n^r r-permutations with repetition.

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Example : How many strings of length r can be formed from the uppercase letters of the English alphabet?

Solution : The number of such strings is 26^r, which is the number of r-permutations of a set with 26 elements.

Dealing with Constraints

Example : How many bit strings of length 10 contain
a) exactly 4 1s? b) At most 3 1s?

a) Solution : To create a bit string with exactly 4 ones, we need to choose 4 positions out of the 10 for the 1s, while the remaining 6 positions will automatically be 0s.
The number of ways to choose 4 positions out of 10 is: $C(10, 4) = 210$.

b) Solution : We need to consider multiple cases:
Case 1: Exactly 0 ones: This corresponds to a string where all 10 bits are 0. There is only 1 such string.
Case 2: Exactly 1 one: Choose 1 position for the 1 out of 10. The number of ways is: $C(10, 1)=10$
Case 3: Exactly 2 ones: Choose 2 positions for the 1s out of 10. The number of ways is: $C(10, 2)=45$
Case 4: Exactly 3 ones: Choose 3 positions for the 1s out of 10. The number of ways is: $C(10, 3)=120$
Total: Now, we use the sum rule for all these values: $1+10+45+120=176$

Dealing with Constraints

Example : 7 female and 9 male faculties are there in the mathematics department of a school. How many ways are there to select a committee of 5 members from the department if at least 1 female faculty is a must on the committee?

Solution : The total number of faculty members is 7 female and 9 male, for a total of $7+9=16$. The number of ways to choose 5 members from $16 = C(16,5) = 4368$.

Now we need to subtract the number of committees that contain no female, or committees that contain only male. Since there are 9 male, the number of ways to choose 5 members from $9 = C(9, 5) = 126$.

Committees with at least 1 female $= 4368-126 = 4242$.

Dealing with Constraints

Example : How many strings of 6 lowercase letters of the English alphabet contain exactly 2 vowels?

Solution : First, we choose 2 positions for the vowels. The number of ways to do this is: $C(6,2) = 15$.

Each vowel position can be filled with any of the 5 vowels. Therefore, there are: $5^2 = 25$ ways to choose the 2 vowels.

The remaining 4 positions must be filled with consonants. Each position can be filled with any of the 21 consonants, so there are: $21^4 = 194\ 481$ ways to choose the 4 consonants.

Finally, we apply product rule to get number of all the strings.
Total strings $= 15 \times 25 \times 194\ 481 = 72\ 930\ 375$

Permutations with Indistinguishable Objects

Example : How many different strings can be made by reordering the letters of the word *SUCCESS*.

Solution : There are seven possible positions for the three Ss, two Cs, one U, and one E.

- The three Ss can be placed in $C(7,3)$ different ways, leaving four positions free.
- The two Cs can be placed in $C(4,2)$ different ways, leaving two positions free.
- The U can be placed in $C(2,1)$ different ways, leaving one position free.
- The E can be placed in $C(1,1)$ way.

By the product rule, the number of different strings is:

$$C(7,3)C(4,2)C(2,1)C(1,1) = \frac{7!}{3!4!} \cdot \frac{4!}{2!2!} \cdot \frac{2!}{1!1!} \cdot \frac{1!}{1!0!} = \frac{7!}{3!2!1!1!} = 420.$$